



M&S
GROUP

EDUCATIONAL CENTER

MECHANICS
OF
MACHINES

midterm

2ش عباس العقاد اعلى بنك الاسكان والتعمير مدينه نصر
126 ش العباسية امام موقف ميدان العباسية – داخل جمعية المواساة الاسلامية



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Past Exams

AIN SHAMS UNIVERSITY
FACULTY OF ENGINEERING
Junior – Mechatronics
Mid Term Examination



Summer 2018 - Semester

Exam Date: 2.8.2018

Exam Time 1:00 Hr.

Theory of machines and multi-body– MCT251

The Exam Consists of Four Questions in Four Pages

Total Marks: 25 Marks

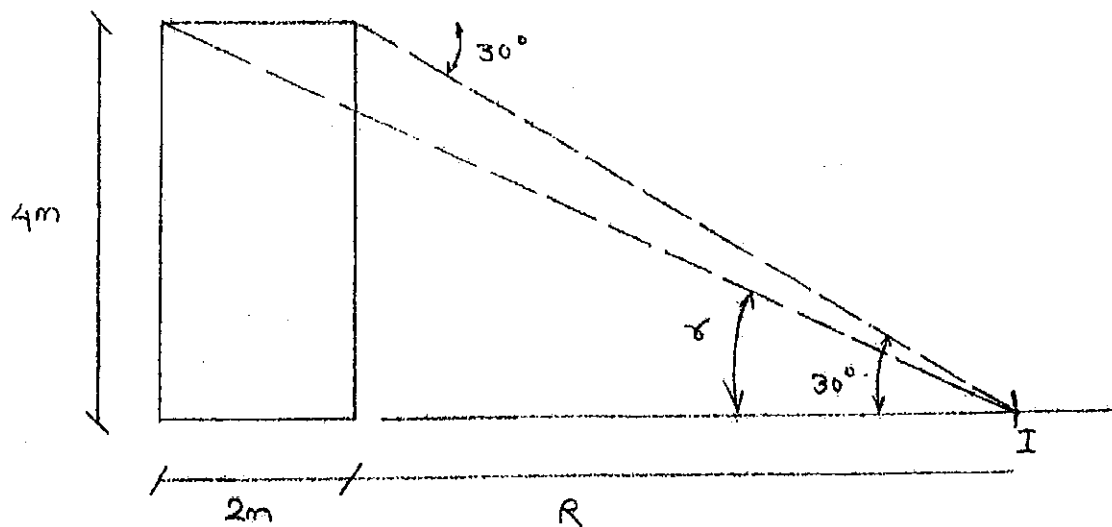
Name:

Sec.: ID:

Question: 1 (4 Marks)

A car is turning a right turn. The distance between the front wheels is 2 m. The distance between the front and rear wheels axes is 4 m. If the right front wheel turns 30 degrees, find, for ideal steering:

- The radius of the turn
- The turning angle of the left front wheels.



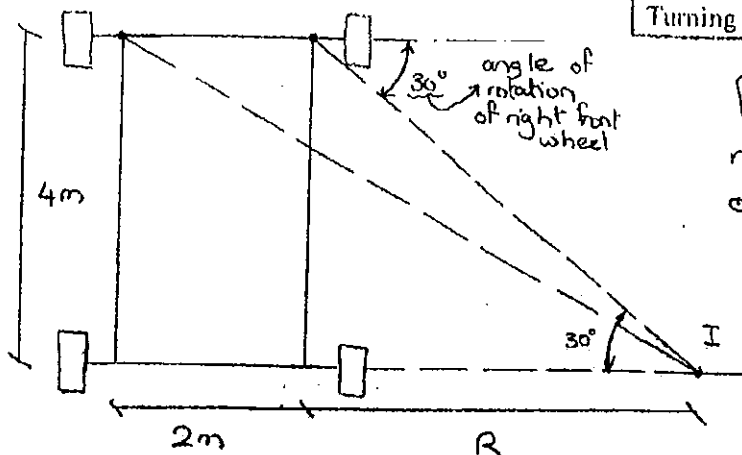
$$a] R = \frac{4}{\tan(30)} = \boxed{6.93 \text{ m}}$$

$$b] \tan \delta = \frac{4}{R+2} = \frac{4}{8.93}$$

$$\Rightarrow \boxed{\delta = 24.1^\circ} \rightarrow \text{turning angle of left front wheel.}$$

Question 2 [2 marks]

A car is turning a right turn. The distance between the front wheels is 2 m. The distance between the front and rear wheels axes is 4 m. If the right front wheel turns 30 degrees. Find, for ideal steering the radius of the turn.



Write Final Answers Here

Turning Radius

6.9 m

For ideal steering, all wheels must have the same center of rotation instantaneously (I)

$$\Rightarrow R = \frac{4}{\tan 30} = 6.9 \text{ m}$$

Question 3 [2 marks]

Two intersecting shafts are connected to transfer motion through a universal joint. If the maximum acceptable speed fluctuation is 20 % of the mean speed.

Find the maximum angle between the axes of the shafts which satisfy this condition.

Write Final Answers Here

Maximum Angle

25.2°

$$\omega_{\max} - \omega_{\min} = 0.2 \omega_{\text{mean}}$$

$$\frac{\phi}{\cos \alpha} - \phi \cos \alpha = 0.2 \left(\frac{\frac{\phi}{\cos \alpha} + \phi \cos \alpha}{2} \right)$$

$$1 - \cos^2 \alpha = 0.1 (1 + \cos^2 \alpha)$$

$$1.1 \cos^2 \alpha = 0.9$$

$$\Rightarrow \alpha = 25.2$$

Question (2): (5 marks)

- a- A car is turning a right turn. The distance between the front wheels is 2 m. The distance between the front and rear wheels axes is 4 m. If the right front wheel turns 30 degrees. Find, for ideal steering the radius of the turn. (3 Marks)

previously

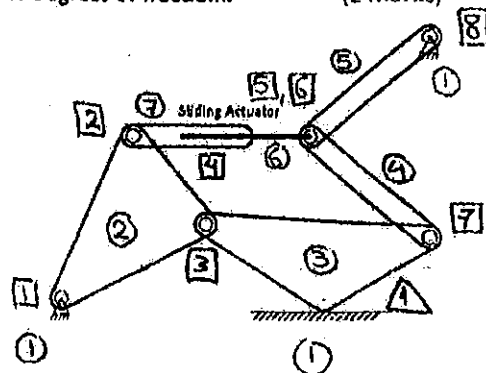
Solved

- b- For the shown linkage, the lower triangular link is kept in contact with ground and the sliding actuator is hinged at each end and its length is changing during actuation. Find the degrees of freedom. (2 Marks)

* $L = 7$

* $P_L = 8$

* $P_h = 1$



$\Rightarrow \text{DOF} = 3L - 2P_L - P_h - 3$

$= 3(7) - 2(8) - 1 - 3 = 1 \text{ dof}$

$\therefore$ constrained mechanism

(34)

الاسم

الفصل رقم الجلوس

AIN SHAMS UNIVERSITY FACULTY OF ENGINEERING

Design and Production Engineering Department
MECHANICAL ENGINEERING PROGRAMS



Summer 2021

Midterm

Time Allowed: 60 min

MDP 212: Mechanics of Machines

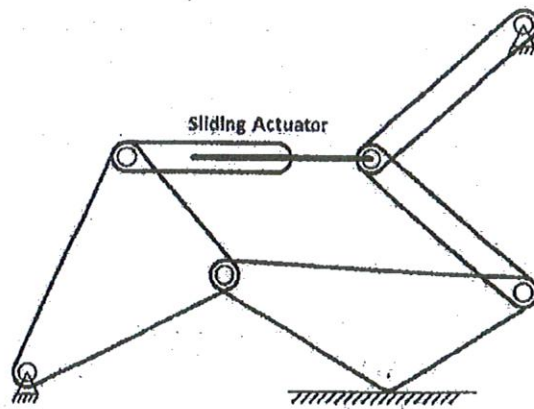
Exam consists of Four question in Four pages.

Total Marks: 15 Marks

1/4

Question (1): [3 Marks]

For the shown linkage, the lower triangular link is kept in contact with ground and the sliding actuator is hinged at each end and its length is changing during actuation. Find the degrees of freedom.



previously
Solved

Name:

Sec.: ID:

AIN SHAMS UNIVERSITY

FACULTY OF ENGINEERING

Junior -Energy & Manufacture Programs

Mid Term Examination



Summer 2019- Semester

Exam Date: 31 - 7 - 2019

Exam Time 1:00Hr.

Theory of machines (MDP254)

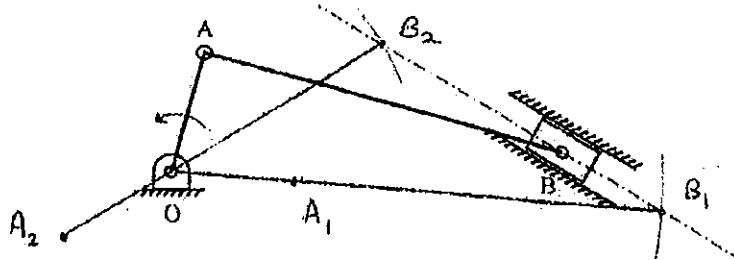
The Exam Consists of Three Questions in Three Pages

Total Marks: 25 Marks

Question: 1 (6 Marks)

The shown mechanism is drawn with its real dimensions. The crank is rotating counter clockwise.

- Draw the mechanism at the two extreme positions.
- Determine the length of the stroke of the block B
- If the mechanism is used for quick return. Determine the angle of rotation of the crank between the two extremes in the forward stroke and the return stroke.
- If the speed of the crank is constant, determine the ratio between the forward and return time of the block B.



b] stroke $B_1 B_2 \approx \boxed{5 \text{ cm}}$

c] Forward stroke " $A_1 A_2$ " $\approx \boxed{216^\circ}$

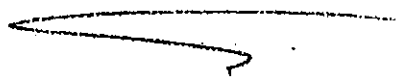
Return stroke " $A_2 A_1$ " $\approx \boxed{144^\circ}$

d] $\frac{t_f}{t_r} = \frac{\theta_f}{\theta_r} = \frac{216}{144} = \boxed{1.5}$

Question: 2 (4 Marks)

Two intersecting shafts are coupled by a Hooke's joint. The driving shaft rotates at a uniform speed 1000 rpm. The total variation in the speed of the driven shaft is not to exceed 50 rpm. Find the greatest permissible angle between the axes of the shafts.

previously solved



AIN SHAMS UNIVERSITY
FACULTY OF ENGINEERING
Design and Production Engineering Department
Junior- Energy
Mid Term Examination



Summer 2020 - Semester

Exam Time 60 Min.

Theory of machines - MDP 254

The Exam Consists of Four Questions in Three Pages

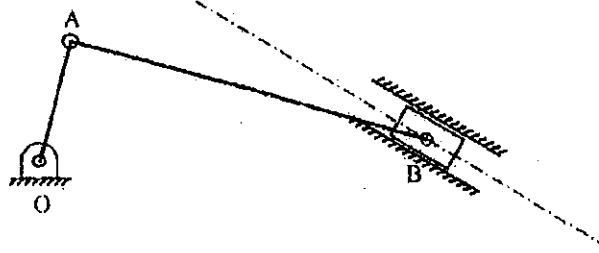
Total Marks: 25 Marks

Answer all Questions

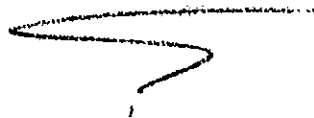
Question: 1 (6 Marks)

The shown mechanism is drawn with its real dimensions. The crank is rotating counter clockwise.

- Draw the mechanism at the two extreme positions.
- Determine the length of the stroke of the block B
- If the mechanism is used for quick return. Determine the angle of rotation of the crank between the two extremes in the forward stroke and the return stroke.
- If the speed of the crank is constant, determining the ratio between the forward and return time of the block B.



previously
Solved



Question: 2 (3 Marks)

Two intersecting shafts are coupled by a Hooke's joint. The driving shaft rotates at a uniform speed 1000 rpm. The total variation in the speed of the driven shaft is not to exceed 50 rpm. Find the greatest permissible angle between the axes of the shafts

previously solved

Midterm Exam

Subject: Theory of machines (MDP254)

Time of exam: 60 minutes

No. of Questions: 4

No. of pages: 4

Name:

Sec:

ID:

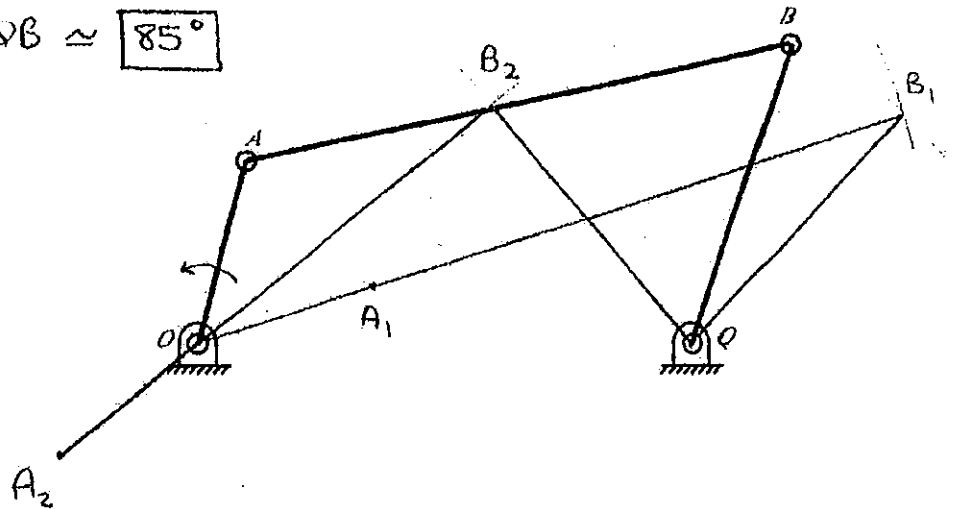
Solve all questions and write the answers and values in the right position (below questions)

Question (1) (6marks)

The shown mechanism is drawn with its real dimensions. The crank OA is rotating counter clockwise. **Find:**

- Draw the mechanism at the two extreme positions.
- Determine the oscillating angle of the link QB
- If the mechanism is used for quick return. Determine the angle of rotation of the crank between the two extremes in the forward stroke and the return stroke.
- If the speed of the crank is constant, determine the ratio between the forward and return time of the lever BQ.

b] Osc. angle of QB $\approx$ 85°



c] Forward stroke $\rightarrow A_1A_2 \rightarrow$ angle of crank $\approx$ 200°
 Return stroke $\rightarrow A_2A_1 \rightarrow$ angle of crank $\approx$ 160°

d]
$$\frac{t_f}{t_r} = \frac{\theta_f}{\theta_r} = \frac{200}{160} = \text{ 1.25 }$$

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AIN SHAMS UNIVERSITY
FACULTY OF ENGINEERING
 Second Year Mechanical Engineering
 MECHANICAL ENGINEERING PROGRAMS



Fall 2018 Midterm-1 Time Allowed: 60 min

MDP 251: Mechanics of Machines (1)

Exam consists of Five questions in four pages.

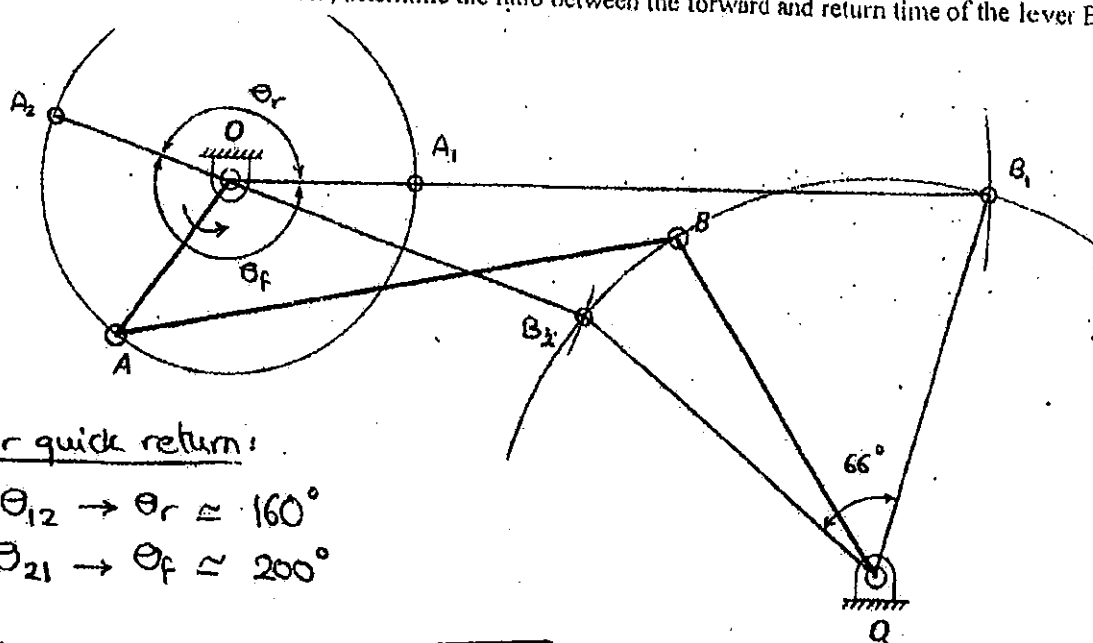
Total Marks: 15 Marks

1/4

Question 1 [3marks]

The shown mechanism is drawn with its real dimensions. The crank OA is rotating counter clockwise.

- Draw the mechanism at the two extreme positions. (Use the shown figure to draw on it)
- Determine the oscillating angle of the link QB
- If the mechanism is used for **quick return**. Determine the angle of rotation of the crank between the two extremes in the **forward stroke** and the **return stroke**.
- If the speed of the crank is constant, determine the ratio between the forward and return time of the lever BQ.



c) For quick return:

$$\theta_{12} \rightarrow \theta_r \approx 160^\circ$$

$$\theta_{21} \rightarrow \theta_f \approx 200^\circ$$

d) $\frac{t_f}{t_r} = \frac{\theta_f}{\theta_r} = \frac{5}{4}$ or $1.25 : 1$

Write Final Answers Here		
b	Oscillating Angle	66°
c	Forward Angle	200°
	Backward Angle	160°
d	Time ratio	1.25 : 1

Name	Sec.	ID.

AIN SHAMS UNIVERSITY
FACULTY OF ENGINEERING
 JUNIOR
 MECHATRONICS AND AUTOMATION ENGINEERING PROGRAM



Summer 2019 Midterm Time Allowed: 80 min

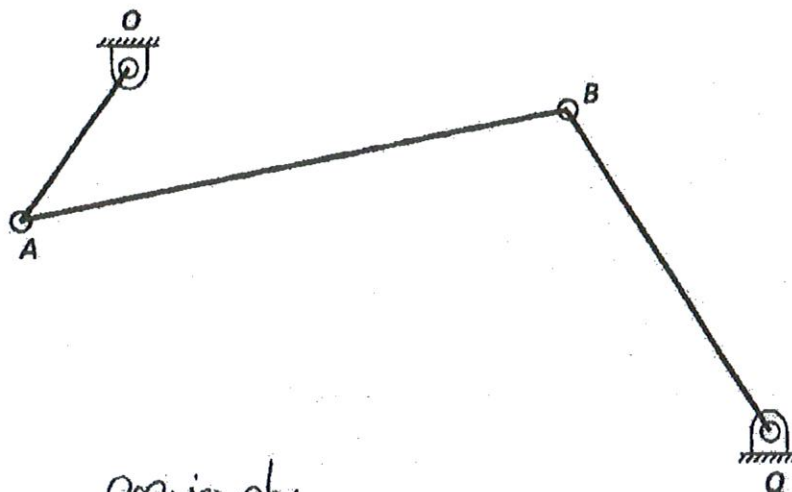
MCT251: Theory of machines and multi-body

Exam consists of Five questions in four pages. Total Marks: 25 Marks 1/4

Question 1 [5 marks]

The shown mechanism is drawn with its **real dimensions**. The crank OA is rotating counter clockwise.

- Draw the mechanism at the two extreme positions. (Use the shown figure to draw on it)
- Determine the oscillating angle of the link QB
- If the mechanism is used for **quick return**. Determine the angle of rotation of the crank between the two extremes in the **forward stroke** and the **return stroke**.
- If the speed of the crank is constant, determine the ratio between the forward and return time of the lever BQ.



*previously
Solved*

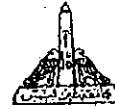
Write Final Answers Here		
b	Oscillating Angle	
c	Forward Angle	
	Backward Angle	
d	Time ratio	

Name:

Sec.: ID:

AIN SHAMS UNIVERSITY
FACULTY OF ENGINEERING

Junior -Energy, Manufacture and Mechatronics
Mid Term Examination



Spring 2021- Semester

Exam Date: 08.05.2021

Exam Time 1:00Hr.

**Theory of machines-MDP254 &
Theory of machines and multi-body- MCT251**

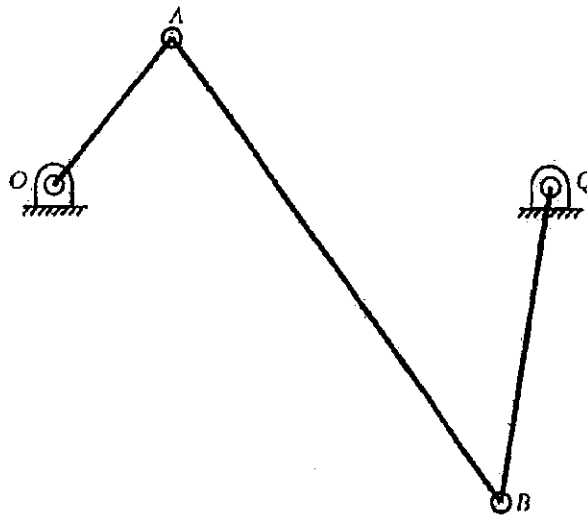
The Exam Consists of Three Questions in Four Pages

Total Marks: 25 Marks

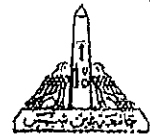
Question: 1 (6 Marks)

The shown mechanism is drawn with its real dimensions. The crank is rotating counterclockwise.

- Draw the mechanism at the two extreme positions.
- Determine the angle of the stroke of the link QB.
- If the mechanism is used for quick return. Determine the angle of rotation of the crank between the two extremes in the forward stroke and the return stroke.
- If the speed of the crank is constant, determine: - The ratio between the forward and return times.



previously
Solved



August 2017, Summer 2017

Time Allowed: 3 Hours

MDP 254: Theory of Machines

Exam consists of Five questions in Five pages.

Total Marks: 40 Marks

1/5

General Instructions:

- Please read the examination paper carefully
- Please assume any missing data in a logical manner.
- Only a non-programmable calculator may be used.

Question (1) (6 Marks)

The tool box of a shaping machine is driven through a crank and slotted lever quick return motion mechanism. If the cutting stroke is to take triple as long as the return stroke. Determine: 1- The relative dimensions of the mechanism. 2- What will be the ratio between the two maxima of the angular velocity of the swinging link?

* for $t_c = 3t_r$

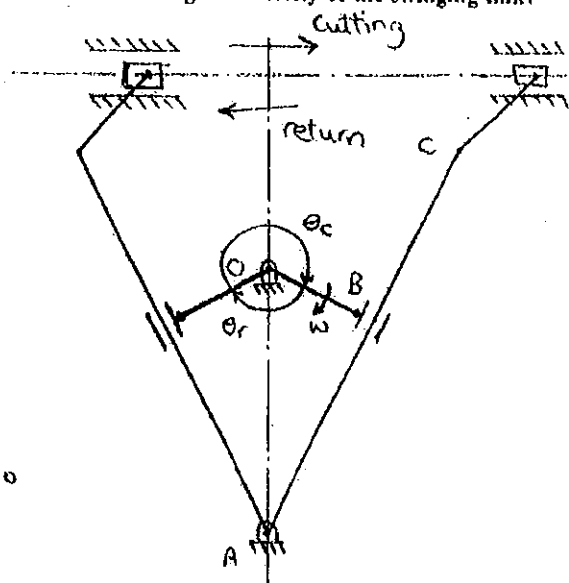
$\Rightarrow \theta_c = 3\theta_r \rightarrow (1)$

* Also we know;

$\theta_c + \theta_r = 360 \rightarrow (2)$

* Solving eqns (1) and (2);

$\Rightarrow \theta_c = 270^\circ \text{ \& } \theta_r = 90^\circ$



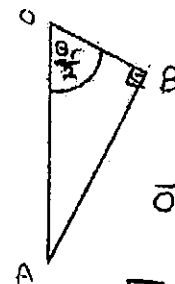
* From $\triangle OAB$;

$\overline{OB} = \overline{OA} \cos \frac{\theta_r}{2}$

$\Rightarrow \overline{OB} = \overline{OA} \cos (45)$

$\frac{\overline{OB}}{\overline{OA}} = \cos (45) = \frac{1}{\sqrt{2}} \rightarrow (1)$

(1)



$\overline{OB}$: crank length

$\overline{OA}$: distance between fixed centers

(9)

Name:

Sec.: ID:

AIN SHAMS UNIVERSITY

FACULTY OF ENGINEERING

Junior --Energy, Manufacture and Mechatronics

Mid Term Examination



Spring 2019- Semester

Exam Date: 2 - 4 - 2019

Exam Time 1:00Hr.

Theory of machines (MDP254 & ERGY342)

Theory of machines and multibody MCT251

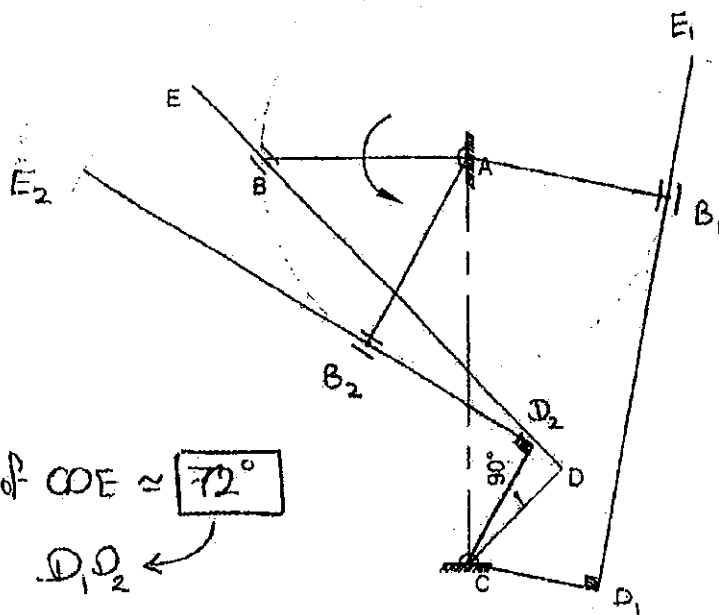
The Exam Consists of Three Questions in Three Pages

Total Marks: 25 Marks

Question: 1 (6 Marks)

The shown mechanism is drawn with its real dimensions. The crank is rotating counter clockwise.

- Draw the mechanism at the two extreme positions.
- Determine the oscillation angle of the link CDE.
- If the mechanism is used for quick return. Determine the angle of rotation of the crank between the two extremes in the forward stroke and the return stroke.
- If the speed of the crank is constant, determine the ratio between the forward and return time of the oscillating link CDE.



b] Osc. angle of CDE $\approx 72^\circ$

$\angle D_1D_2 \leftarrow$

c] forward stroke " B_1B_2 " = 252°

Return stroke " B_2B_1 " = 108°

d] $\frac{t_f}{t_r} = \frac{\theta_f}{\theta_r} = \frac{252}{108} = \frac{7}{3}$

Question: 3 (4 Marks)

Two shafts A and B are connected by Hooke's joint with axes inclined at an angle α . If the driving shaft A rotate 15° from the first position at which the two shafts speed are equal. Find the speed of the driven shaft B at that instant, if $\alpha = 15^\circ$, and the speed of shaft A is $\omega = 60$ rad/s.

List of Equations: -

$$\cos 2\theta \cong \frac{2 \sin^2 \alpha}{2 - \sin^2 \alpha}$$

$$\tan \theta = \pm \sqrt{\cos \alpha}$$

$$\omega_1 = \frac{-\omega^2 \cos \alpha \sin^2 \alpha \sin 2\theta}{(1 - \cos^2 \theta \sin^2 \alpha)^2}$$

Solution:

$$\begin{aligned} * \text{ at } \omega = \omega_1 &\rightarrow \tan \theta = + \sqrt{\cos \alpha} = \sqrt{\cos(15)} \\ &\Rightarrow \theta = 44.5^\circ \end{aligned}$$

* For this instant ;

$$\theta = 44.5 + 15 = 59.5^\circ$$

$$\frac{\omega}{\omega_1} = \frac{1 - \cos^2 \theta \sin^2 \alpha}{\cos \alpha}$$

$$\begin{aligned} \Rightarrow \omega_1 &= \frac{\omega \cos \alpha}{1 - \cos^2 \theta \sin^2 \alpha} = \frac{60 \cos(15)}{1 - \cos^2(59.5) \sin^2(15)} \\ &= \boxed{59 \text{ rad/s}} \end{aligned}$$

مع أطيب أمنياتي بالتوفيق والنجاح

Examiner: Associate Professor Hossam Dogheim

Name: _____

Sec.: _____

ID: _____

Question: 3 (4 Marks)

Two shafts A and B are connected by Hooke's joint with axes inclined at an angle α . If the driving shaft A rotate 15° from the first position at which the two shafts speed are equal. Find the speed of the driven shaft B at that instant, if $\alpha = 15^\circ$, and the speed of shaft A is $\omega = 60$ rad/s.

List of Equations: -

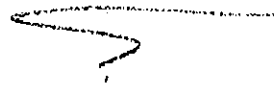
$$\cos 2\theta \cong \frac{2 \sin^2 \alpha}{2 - \sin^2 \alpha}$$

$$\tan \theta = \pm \sqrt{\cos \alpha}$$

$$\omega_1 = \frac{-\omega^2 \cos \alpha \sin^2 \alpha \sin 2\theta}{(1 - \cos^2 \theta \sin^2 \alpha)^2}$$

previously

Solved



مع أطيب أمنياتنا بالتوفيق والنجاح

Examiner: Associate Professor Hassam Doghiem

Question (3) (4marks)

Two intersecting shafts are coupled by a Hooke's joint. The driving shaft rotates at a uniform speed 2000 rpm. The total variation in the speed of the driven shaft is not to exceed 100 rpm.

Find: the greatest permissible angle between the axes of the shafts.

$$* \quad \omega = 2000 \text{ rpm}$$

$$* \quad \omega_{\max} - \omega_{\min} = 100 \text{ rpm}$$

$$* \quad \alpha = ?$$

Solution:

$$\rightarrow \omega_{\max} - \omega_{\min} = \frac{\omega}{\cos \alpha} - \omega \cos \alpha = 100$$

$$\Rightarrow \omega \left(\frac{1 - \cos^2 \alpha}{\cos \alpha} \right) = 100$$

given $\omega = 2000 \text{ rpm}$ & solving for α ;

$$\Rightarrow \alpha = \boxed{12.76^\circ}$$

(30)

Name	Group	Sec.	ID

AIN SHAMS UNIVERSITY
FACULTY OF ENGINEERING
CREDIT HOURS ENGINEERING PROGRAMS
Mechatronics and Automation Engineering Program



Spring 2021

Course Code: MDP 212

Time allowed: 1:00 Hrs.

Mechanics of Machines – Midterm Exam

The Exam Consists of Three Questions in Four Pages.

Maximum Marks: 25 Marks

1 / 4

Important Rules:

- Having a "turned ON" mobiles inside the examination hall is forbidden and is considered as a cheating behaviour. If you should have your mobile with you, it must be turned off and placed into your own bag.
- Any kind of "Smart Watches" or earphones (Bluetooth, wireless, or wired) are forbidden.
- It is forbidden to have any references, notes, books, or any other materials even if it is not related to the exam content with you in the examination hall.
- Clarify your answer with all data, dimensions, sketches, and annotations. In addition, assume logically any missing data.

تعليمات هامة

- حيازة التليفون المحمول مفتوحا داخل لجنة الامتحان يعتبر حالة غش تستوجب العقاب وإذا كان ضروري الدخول بالمحمول فيوضع مغلق في الحقيبة.
- لا يسمح بدخول الساعات الذكية ولا سماعة الأذن أو البلوتوث.
- لا يسمح بدخول أي كتب أو ملازم أو أوراق داخل اللجنة والمخالفة تعتبر حالة غش.
- وضع جميع إجاباتك بذكر كافة المعلومات والأبعاد اللازمة، والفترض بمنطقية أية أبعاد غير مذكورة صراحة.

Question (1): (5 points)

Two intersecting shafts are connected to transfer motion through a universal joint. If the maximum acceptable speed fluctuation is 20 % of the mean speed. Find the maximum angle between the axes of the shafts which satisfy this condition.

previously
Solved

Question 2 [3 marks]

Two intersecting shafts are connected to transfer motion through a universal joint. If the maximum acceptable speed fluctuation is 20 % of the mean speed. Find the maximum angle between the axes of the shafts which satisfy this condition.

Write Final Answers Here

Maximum Angle

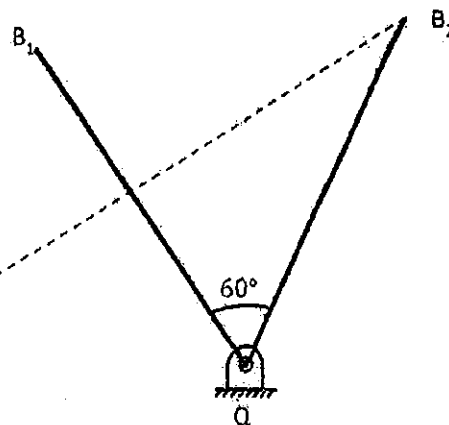
previously
solved

Question 3 [5 marks]

It is required to design a crank-rocker four-bar mechanism QABQ. OA is the crank, AB is the coupler, QB is the rocker, QQ is the fixed link. The link QB is shown in the figure with its real length. It is required to oscillate the link QB 60° between the shown positions. The time to move from position (1) to position (2) is equal to 0.8 of the time to move from the position (2) to position (1).

If the hinge Q is on the shown dashed line.

- Show the proper position of the turning pair (hinge) O.
- Determine the length of the crank OA.
- Determine the length of the coupler AB.
- Determine the length of the fixed link QQ.



Write Final Answers Here

b	Length of OA	
c	Length of AB	
d	Length of QQ	

(22)

Question: 2 (4 Marks)

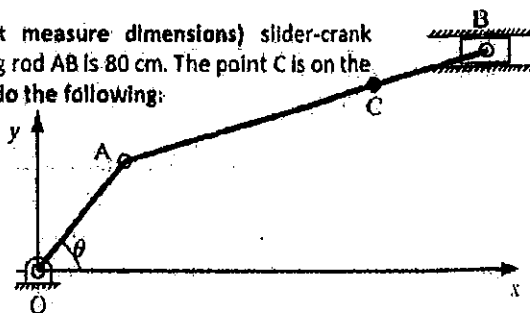
Two intersecting shafts are coupled by a Hooke's joint. The driving shaft rotates at a uniform speed 2000 rpm. The total variation in the speed of the driven shaft is not to exceed 100 rpm. Find the greatest permissible angle between the axes of the shafts.

previously solved

Question: 3 (9 Marks)

The following figure represents a schematic (don't measure dimensions) slider-crank mechanism OAB. The crank OA is 30 cm. The connecting rod AB is 80 cm. The point C is on the link AB and AC is 50 cm. When the angle θ equals 60° , do the following:

- Show the vector loop on the mechanism.
- Write the loop equation.
- Solve the loop equation.
Hence, determine analytically
- The position of block B. (x and y coordinates)
- The of the point C. (x and y coordinates)



Question 4 [4 marks]

It is required to design a crank-rocker four-bar mechanism OABQ. OA is the crank. AB is the coupler. QB is the rocker. OQ is the fixed link. The link QB is shown in the figure with its real length. It is required to oscillate the link QB 60° , between the shown positions. The time to move from position (1) to position (2) is equal to 0.8 of the time to move from the position (2) to position (1).

If the hinge O is on the shown dashed line.

- Show the proper position of the turning pair (hinge) O.
- Determine the length of the crank OA.
- Determine the length of the coupler AB.
- Determine the length of the fixed link OQ.

$$t_{12} = 0.8 t_{21}$$

$$\theta_{12} = 0.8 \theta_{21} \rightarrow \textcircled{1}$$

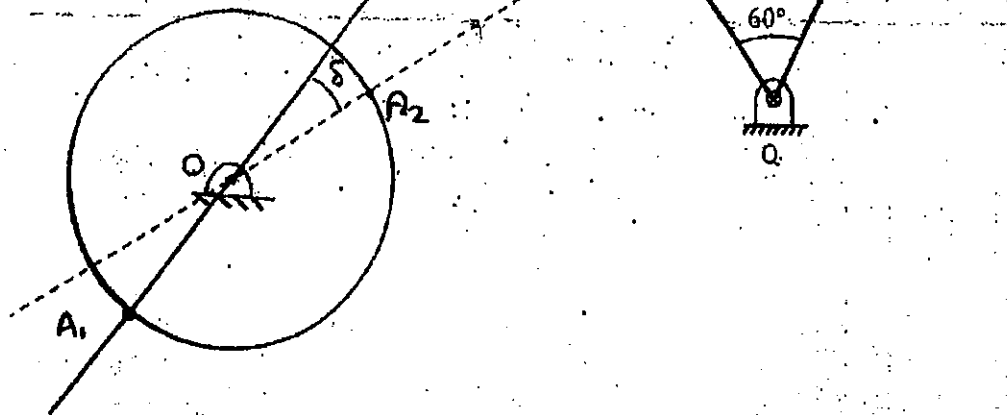
$$\text{Also ; } \theta_{12} + \theta_{21} = 360 \rightarrow \textcircled{2}$$

$$\Rightarrow \theta_{12} = 160, \theta_{21} = 200$$

$$\therefore \delta = \frac{\theta_{21} - \theta_{12}}{2} = 20^\circ$$

From Fig $OB_2 = 12.7 \text{ cm}$
 $OB_1 = 7.7 \text{ cm}$

$$\therefore OA = \frac{12.7 - 7.7}{2} = 2.5 \text{ cm}$$



$$\therefore AB = 10.2 \text{ cm} \quad \& \quad OQ = 8.35 \text{ cm}$$

Write Final Answers Here

b	Length of OA	2.5 cm
c	Length of AB	10.2 cm
d	Length of OQ	8.4 cm

AIN SHAMS UNIVERSITY
FACULTY OF ENGINEERING
 Design and Production Engineering Department
MECHANICAL ENGINEERING PROGRAMS



Fall 2020

Midterm

Time Allowed: 50 min

MDP 212: Mechanics of Machines

Exam consists of Three questions in Three pages.

Total Marks: 15 Marks

1/3

Question (1): [5 Marks]

It is required to design a crank-rocker four-bar mechanism OABQ. OA is the crank, AB is the coupler, QB is the rocker, OQ is the fixed link. The link QB is shown in Figure.1 with its real length. It is required to oscillate the link QB 60° between the shown positions as in Figure.1. The time to move from position (1) to position (2) is equal to 0.8 of the time to move from the position (2) to position (1). Do the following:

- Show the position of turning pair (hinge) O, which satisfies the condition.
- Determine the length of the crank OA.
- Draw the mechanism in its extreme positions.
- Determine the length of the coupler AB.

Previously
Solved

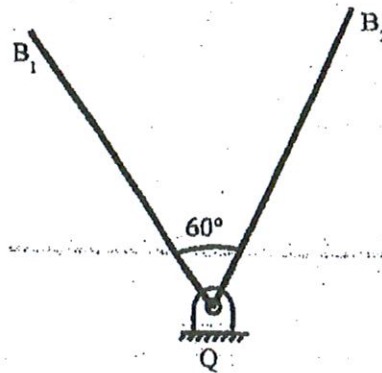


Figure. 1

Question (3): (7 points)

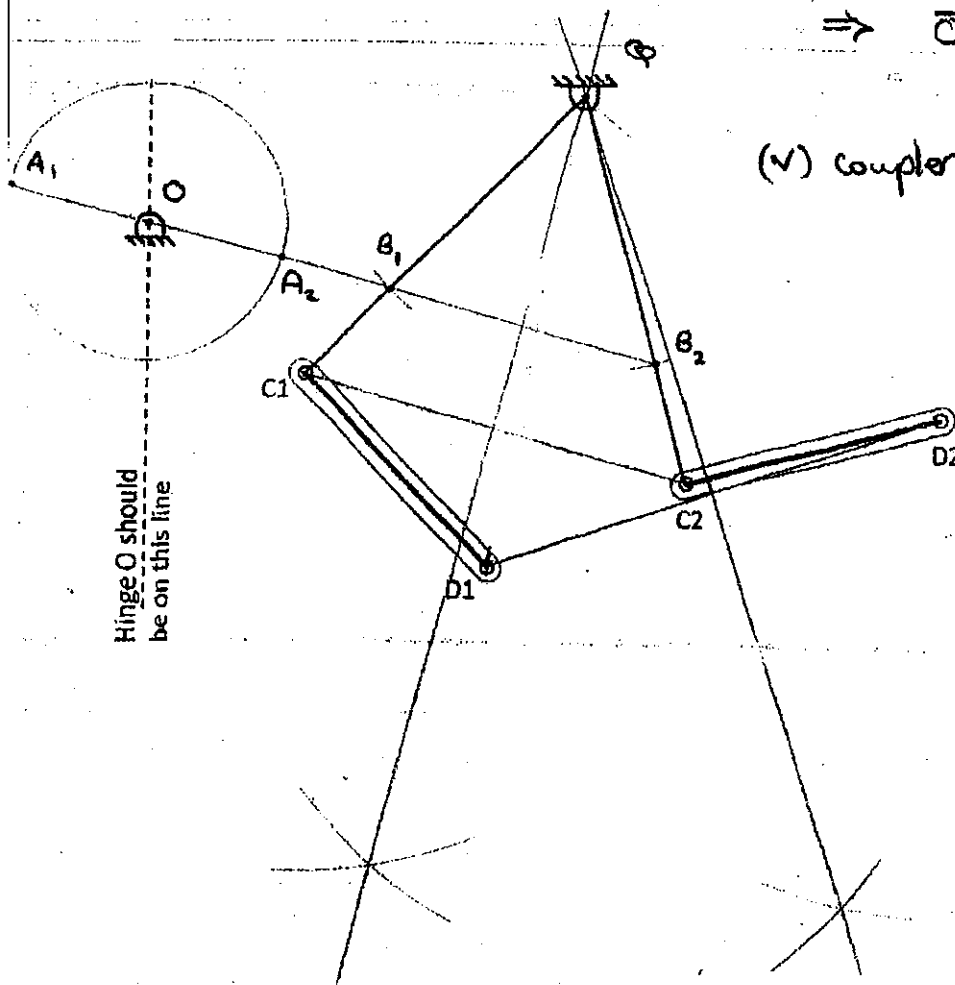
It is required to design a crank-rocker four-bar mechanism to move the link CD as shown in the figure with its real length from the shown positions. The link CD is compound with rocker to form one rigid link QBCD. QBCD is a bell crank. The four-bar mechanism is OABQ. OA is the crank. AB is the coupler. QBCD is the rocker. The OQ is the fixed link. The forward and return times are equal. The hinge O is on the shown dashed line. Consider QB equal 4cm.

- Show the proper position of the turning pair (hinge) Q.
- Draw the link QBCD.
- Show the proper position of the turning pair (hinge) O.
- Determine the length of the crank OA.
- Determine the length of the coupler AB.

(iv) By bisecting $\overline{B_1B_2}$

$$\Rightarrow \overline{OA} = \frac{\overline{B_1B_2}}{2} = \frac{4}{2} = \boxed{2\text{cm}}$$

(v) coupler $\overline{AB} \approx \boxed{5.7\text{cm}}$

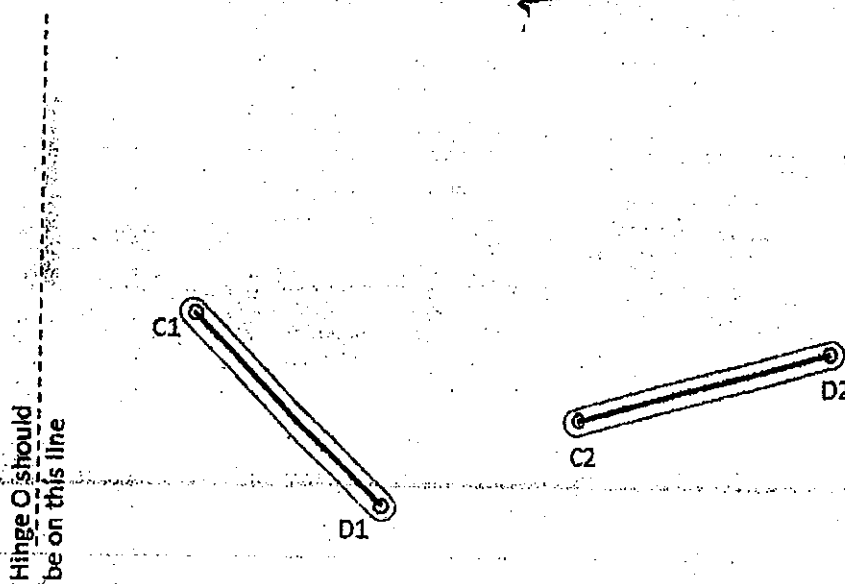


Question (4): [4 marks]

It is required to design a crank-rocker four-bar mechanism to move the link CD as shown in the figure with its real length from the shown positions. The link CD is compound with rocker to form one rigid link QBCD. QBCD is a bell crank. The four-bar mechanism is OABQ. OA is the crank. AB is the coupler. QBCD is the rocker. The OQ is the fixed link. The forward and return times are equal. The hinge Q is on the shown dashed line. Consider QB equal 4cm.

- Show the proper position of the turning pair (hinge) Q.
- Draw the link QBCD.
- Show the proper position of the turning pair (hinge) O.
- Determine the length of the crank OA.
- Determine the length of the coupler AB.

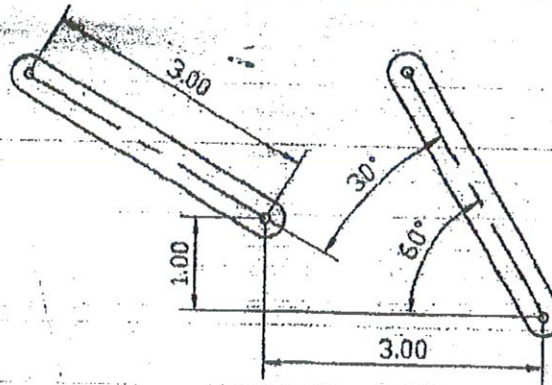
*Previously
Solved*



Student Name: ID:

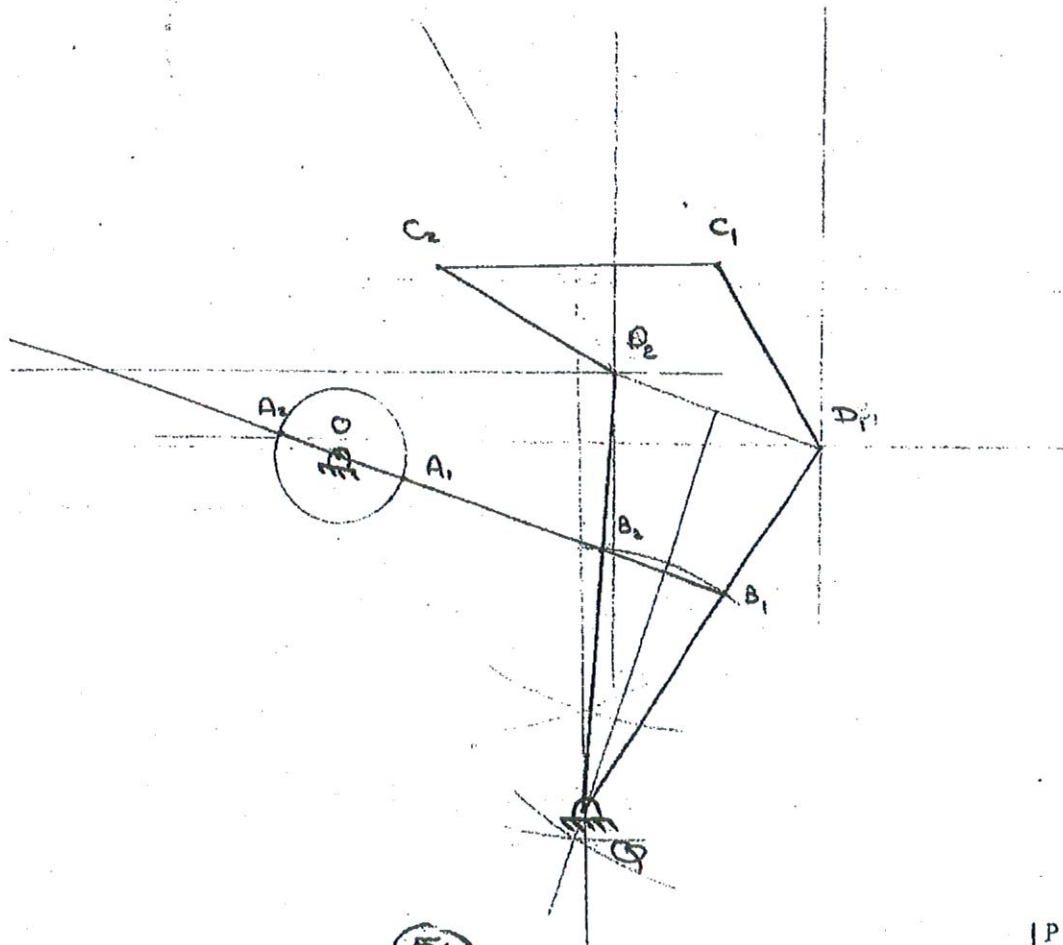
Question (2): [8 Marks]

Design graphically a Grashof four bar mechanism with rocker motion reciprocating between the 2 shown positions. The given drawing is not to scale and all dimensions are in cm.



Answer:

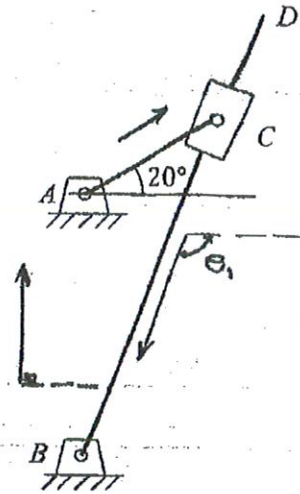
$$* \text{Crank } \overline{OA} = \frac{B_1B_2}{2} = \frac{1.9}{2} = \boxed{0.95 \text{ cm}}$$



(54)

Question (2): [4 marks]

The shown mechanism consists of hinged lever AD and crank AC through a block C a telescopic link BD. AB = 7 cm, AC = 4 cm. Find the positions of block C and the link BD.



* Loop ACBA:

$$\vec{R}_{CA} + \vec{R}_{BC} + \vec{R}_{AB} = 0$$

$$4 \angle 20 + \overline{BC} \angle \theta_1 + 7 \angle 90 = 0$$

* hor-comp.

$$4 \cos(20) + \overline{BC} \cos \theta_1 + 7 \cos(90) = 0$$

$$\therefore \overline{BC} \cos \theta_1 = -3.76 \rightarrow \textcircled{1}$$

* vert. comp.

$$4 \sin(20) + \overline{BC} \sin \theta_1 + 7 \sin(90) = 0$$

$$\therefore \overline{BC} \sin \theta_1 = -8.37 \rightarrow \textcircled{2}$$

* $\textcircled{2} / \textcircled{1}$:

$$\therefore \tan \theta_1 = 2.23 \Rightarrow \theta_1 = 65.8^\circ \text{ or } \theta_1 = 245.8^\circ$$

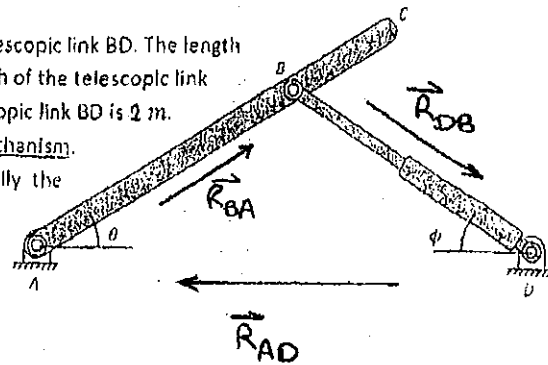
* $\textcircled{1}^2 + \textcircled{2}^2$:

$$\therefore \overline{BC}^2 = 84.2 \Rightarrow \overline{BC} = 9.2 \text{ cm}$$

Question (2): [7 marks]

The shown mechanism consists of hinged lever ABC and a telescopic link BD. The length of the fixed link AD is 5 m. The length of AB is 4 m. The length of the telescopic link is changeable. At the shown position the length of the telescopic link BD is 2 m.

- Write the loop equation, show the vector loop on the mechanism.
- Hence, solve the loop equation to determine analytically the angles θ and the angle ϕ at this position.



Loop ABDA:

$$\vec{R}_{BA} + \vec{R}_{DB} + \vec{R}_{AD} = 0$$

$$4 \angle \theta + 2 \angle \phi + 5 \angle 180 = 0$$

Using Euler's form;

$$\Rightarrow 4e^{i\theta} + 2e^{i\phi} + 5e^{i(180)} = 0$$

$$4e^{i\theta} + 2e^{i\phi} = -5e^{i(180)}$$

$$\textcircled{1} \text{ let } s = e^{i\theta}, t = e^{i\phi}, z = -5e^{i(180)}$$

$$\Rightarrow 4s + 2t = z \rightarrow \textcircled{1}$$

$$\text{but } z = -5e^{i(180)} = -5 \cos(180) - 5i \sin(180)$$

$$= 5 \quad \bar{z} = 5$$

$\textcircled{2}$ Find the conjugate of eqs $\textcircled{1}$;

$$\frac{4}{s} + \frac{2}{t} = \bar{z} \rightarrow \textcircled{2}$$

(7)

③ Solve eqns ① & ②,

From ①; $s = \frac{z-2t}{4}$

In ②; $\frac{4}{\frac{z-2t}{4}} + \frac{2}{t} = \bar{z}$

$$\frac{16}{z-2t} + \frac{2}{t} = \bar{z}$$

$$16t + 2(z-2t) = \bar{z}t(z-2t)$$

$$16t + 2z - 4t = z\bar{z}t - 2\bar{z}t^2$$

$$12t + 10 = 25t - 10t^2$$

$$10t^2 - 13t + 10 = 0$$

④ Solve for t:

$$A = 10 + 0i, \quad B = -13 + 0i, \quad C = 10 + 0i$$

Using formula; $t = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$

$$\Rightarrow t = 0.65 + 0.76i \quad \text{OR} \quad \underline{t = 0.65 - 0.76i}$$

Accepted vector

$$\Rightarrow \tan \phi = \frac{-0.76}{0.65} \Rightarrow \phi = \underline{-49.5}$$

Accepted angle

OR
130.5°

(51)

$$\rightarrow t = 0.65 - 0.76i$$

$$\Rightarrow S = \frac{1}{4} (\bar{z} - 2(0.65 - 0.76i))$$

$$= 0.925 + 0.38i$$

$$\Rightarrow \tan \theta = \frac{0.38}{0.925}$$

$$\rightarrow \theta = 22.3^\circ$$

OR

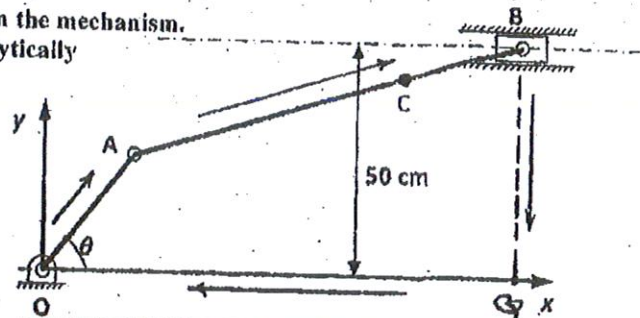
202.3

Accepted angle.

Question 5 [4 marks]

The following figure represents a schematic (don't measure dimensions) slider-crank mechanism OAB. The crank OA is 30 cm. The connecting rod AB is 80 cm. The point C is on the link AB and AC is 50 cm. When the angle θ equals 60° . Do the following:

- Write the loop equation, show the vector loop on the mechanism. Solve the loop equation. Hence, determine analytically
- The position of block B. (x and y coordinates)
- The of the point C. (x and y coordinates)



Answer should be using loop equation method
any other solutions will not be marked

a) Loop OABCO:

$$\vec{r}_{AO} + \vec{r}_{BA} + \vec{r}_{CB} + \vec{r}_{BO} = 0$$

$$30 \angle 60 + 80 \angle \theta_1 + 50 \angle -90 + 0 \angle 180 = 0$$

Re parts:

$$30 \cos(60) + 80 \cos \theta_1 + 50 \cos(-90) + 0 \cos(180) = 0 \quad \rightarrow (1)$$

Im parts:

$$30 \sin(60) + 80 \sin \theta_1 + 50 \sin(-90) + 0 \sin(180) = 0$$

$\Rightarrow$ Solving for $\theta_1 \rightarrow \theta_1 = 17.5^\circ$ or 162.5°
accepted value $\leftarrow$ refused

In eqn (1):

$$30 \cos(60) + 80 \cos(17.5) + 50 \cos(-90) + 0 \cos(180) = 0$$

$\Rightarrow$ solving for $OB \rightarrow OB = 91.3 \text{ cm}$ # (b)

c) $x_C = OA \cos 60 + AC \cos \theta_1 = 62.7 \text{ cm}$

$y_C = OA \sin 60 + AC \sin \theta_1 = 41 \text{ cm}$

OR

Solve loop eqn. OACO
then resolve $\vec{r}_{OC}$

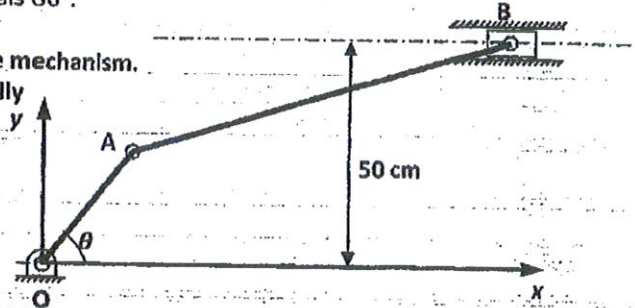
Write Final Answers Here		
b	x_B	91.3 cm
	y_B	50 cm
c	x_C	62.7 cm
	y_C	41 cm

Question 4 [6 marks]

The following figure represents a schematic (don't measure dimensions) slider-crank mechanism OAB. The crank OA is 30 cm. The connecting rod AB is 80 cm. When the angle θ equals 60° .

Do the following:

- Write the loop equation, show the vector loop on the mechanism. Solve the loop equation. Hence, determine analytically
- The position of block B. (x and y coordinates)



Answer should be using loop equation method
any other solutions will not be marked

Previously
Solved

Write Final Answers Here		
b	x_B	
	y_B	

Review

Quiz Problem

The given figure shows a mechanism. Point C is a point on the frame which is adjacent to point B. The frame (containing point C) is moving with velocity 16 cm/s downwards and acceleration 10 cm/s² downwards. The mass of the frame is 10 kg. Find:

- The angular velocity of the link AB (value and direction).
- The angular acceleration of link AB (value and direction).
- The torque about A required to move the frame.

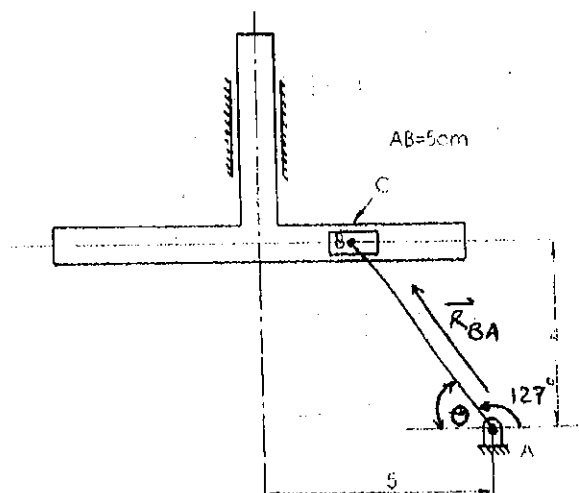
→ Neglect gravitational forces

Givens:

$$v_c = 16 \text{ cm/s} \downarrow$$

$$a_c = 10 \text{ cm/s}^2 \downarrow$$

$$m_f = 10 \text{ kg}$$



$$\tan \theta = \frac{4}{3} \Rightarrow \theta = 53^\circ$$

Solution:

Assume point "D" on the fixation below points "B" & "C".

* Loop ABCDA:

$$\vec{v}_{BA} + \vec{v}_{CB} + \vec{v}_{DC} + \vec{v}_{AD} = 0$$

$$AB \omega_{AB} \begin{matrix} 127 \\ +90 \end{matrix} + v_{CB}^s \begin{matrix} 0 \\ 0 \end{matrix} + v_{DC}^s \begin{matrix} +90 \\ 0 \end{matrix} = 0$$

$$5 \omega_{AB} \begin{matrix} 217 \\ 0 \end{matrix} + v_{CB}^s \begin{matrix} 0 \\ 0 \end{matrix} + 16 \begin{matrix} +90 \\ 0 \end{matrix} = 0$$

Re parts:

$$5 \omega_{AB} \cos(217) + v_{CB}^s \cos(0) + 16 \cos(90) = 0$$

$$-4 \omega_{AB} + v_{CB}^s = 0 \rightarrow (1)$$

Im parts:

$$5 \omega_{AB} \sin(217) + v_{CB}^s \sin(0) + 16 \sin(90) = 0$$

$$\Rightarrow \boxed{\omega_{AB} = 5.3 \text{ rad/s} \curvearrowright} \quad \# (2)$$

MDP 254: Theory of Machines

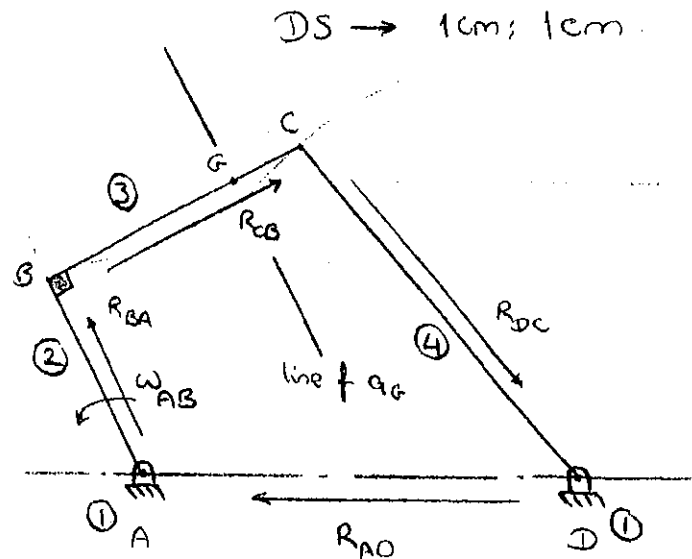
Exam consists of Five questions in Five pages.

3/5

Question (3) (12 Marks)

In the four-bar chain ABCD, A and D are fixed centers 6.0 cm apart on a horizontal line. The driving crank AB = 3 cm, the driven crank DC = 6 cm and coupler BC = 4 cm with a point (G) at 1 cm from C. when the angle ABC is 90 deg. and, B and C are on the same side of AD. If for this position, the angular velocity of AB is 10 rad./sec. anti-clockwise, find the angular velocities of BC and DC and the linear velocity of point G. (value and direction). If a mass of 10 Kg is fixed at point G find the torque required on AB to accelerate the mass at G. (neglect gravity effect).

- $\omega_{AB} = 10 \text{ rad/s } \uparrow$
- $\overline{AB} = 3 \text{ cm} \quad \overline{AD} = 6 \text{ cm}$
- $\overline{DC} = 6 \text{ cm}$
- $\overline{BC} = 4 \text{ cm}$
- $\overline{CG} = 1 \text{ cm}$

Velocity Analysis :Loop ABCDA :

$$\vec{R}_{BA} + \vec{R}_{CB} + \vec{R}_{DC} + \vec{R}_{AD} = 0$$

$$\vec{V}_{BA} + \vec{V}_{CB} + \vec{V}_{DC} + \vec{V}_{AD} = 0$$

$$\overline{AB} \omega_{AB} \begin{matrix} 119 \\ +90 \end{matrix} + \overline{BC} \omega_{BC} \begin{matrix} 29 \\ +90 \end{matrix} + \overline{CD} \omega_{DC} \begin{matrix} -49 \\ +90 \end{matrix} = 0$$

Re parts :

$$3(10) \cos(209) + 4 \omega_{BC} \cos(119) + 6 \omega_{DC} \cos(41) = 0$$

$$-1.9 \omega_{BC} + 4.5 \omega_{DC} = 26.2 \rightarrow (1)$$

Im parts :

$$3(10) \sin(209) + 4 \omega_{BC} \sin(119) + 6 \omega_{DC} \sin(41) = 0$$

$$3.5 \omega_{BC} + 3.9 \omega_{DC} = 14.5 \rightarrow (2)$$

Cont. Q 3:

Solving eqns ① & ②:

$$\omega_{BC} = -1.6 \text{ rad/s} = \boxed{1.6 \text{ rad/s CW}}$$

$$\omega_{DC} = \boxed{5.1 \text{ rad/s CCW}}$$

Loop ABGA:

$$\vec{V}_{BA} + \vec{V}_{GB} + \vec{V}_{AG} = 0$$

$$\overline{AB} \omega_{AB} \underline{209} + \overline{BG} \omega_{BC} \underline{119} + V_{AG} \underline{\theta} = 0$$

Re parts:

$$3(10) \cos(209) + 3(-1.6) \cos(119) + V_{AG} \cos \theta = 0$$

$$\Rightarrow V_{AG} \cos \theta = 23.9 \rightarrow \text{①}$$

Im parts:

$$3(10) \sin(209) + 3(-1.6) \sin(119) + V_{AG} \sin \theta = 0$$

$$\Rightarrow V_{AG} \sin \theta = 18.7 \rightarrow \text{②}$$

$$\ast \frac{\text{②}}{\text{①}}; \tan \theta = 0.78 \Rightarrow \boxed{\theta = 38.1^\circ}$$

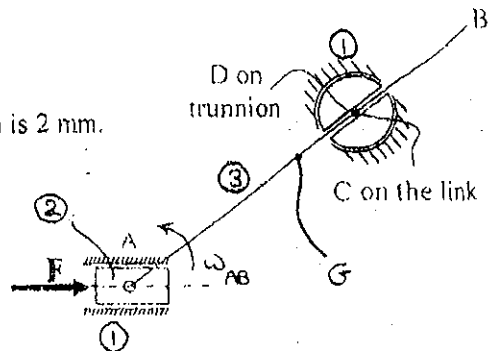
$$\ast \text{①}^2 + \text{②}^2; V_{AG}^2 = 920.9 \Rightarrow \boxed{V_{AG} = 30.3 \text{ cm/s}}$$

$$\therefore \boxed{\vec{V}_{GA} = 30.3 \underline{-141.9}}$$

Question (7): (35 marks)

The shown mechanism is drawn with its **real dimensions**. The mechanism is moving in **horizontal plane**. The link AB is 60 mm. The length of distance AC is 40 mm. The link AB is uniform with mass center at its midpoint. The mass of link AB is 0.4 kg with 20 mm radius of gyration. The mass of the block A is 0.1 kg. Assume no friction in the turning and sliding pairs. The link AB is rotating with constant angular velocity 1 rad/s C.C.W. For the shown position find:

- The velocity of the block A.
- The sliding velocity of link AB relative to point D.
- The rubbing velocity of the pin A, where the radius of the pin is 2 mm.
- The acceleration of the block A.
- The force **F** required to drive the mechanism.
- Find the normal force on the block A.
- Find the normal force between the link AB and the trunnion.



Solution :

Velocity Analysis :

Loop OACDO :

$$\vec{V}_{AO} + \vec{V}_{CA} + \vec{V}_{DC} + \vec{V}_{DO} = 0$$

$$V_{AO}^s \angle 180 + AC \omega_{AB} \angle 36 + 90 + V_{DC}^s \angle 36 = 0$$

$$V_{AO}^s \angle 180 + 40 (1) \angle 126 + V_{DC}^s \angle 36 = 0$$

In parts :

$$0 + 0.59 V_{DC}^s = -32.4 \rightarrow \textcircled{1}$$

$$\therefore V_{DC}^s = -54.8 \text{ mm/s}$$

$$\therefore \vec{V}_{DC} = 54.8 \angle -144 \text{ mm/s}$$

OR $\vec{V}_{CD} = 54.8 \angle 36 \text{ mm/s}$ # $\textcircled{b}$

Cont. Q7:

Re parts:

$$-V_{AO}^s = 67.8 \rightarrow V_{AO}^s = -67.8 \text{ mm/s}$$

$$\therefore \vec{V}_{AO} = \boxed{67.8 \mid 0 \text{ mm/s}} \quad \# \text{ (a)}$$

$$\begin{aligned} \text{c) } V_{AB} &= \omega_{AB} r_A \\ &= 1(2) = \boxed{2 \text{ mm/s}} \quad \# \text{ (c)} \end{aligned}$$

Acceleration Analysis:

Loop OACDO:

$$\vec{a}_{AO} + \vec{a}_{CA} + \vec{a}_{DC} + \vec{a}_{DO} = 0$$

$$a_{AO}^s \mid 180 + AC \omega_{AB}^2 \mid 36+180 + 2\omega_{AB} V_{DC}^s \mid 36+90 + a_{DC}^s \mid 36 = 0$$

$$a_{AO}^s \mid 180 + 40(1)^2 \mid 210 + 2(1)(-54.8) \mid 126 + a_{DC}^s \mid 36 = 0$$

In parts:

$$0 + 0.59 a_{DC}^s = 108.7$$

$$\Rightarrow a_{DC}^s = 184.2 \text{ mm/s}^2$$

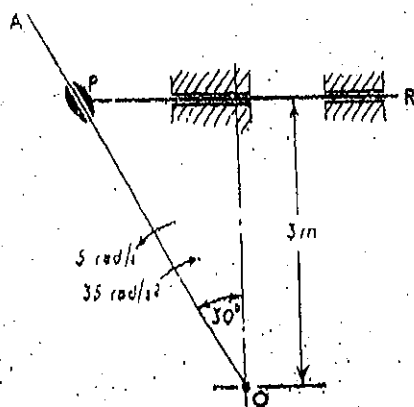
Re parts:

$$-a_{AO}^s = -178.8 \Rightarrow a_{AO}^s = 178.8 \text{ mm/s}^2$$

$$\Rightarrow \vec{a}_{AO} = \boxed{178.8 \mid 180 \text{ mm/s}^2} \quad \# \text{ (d)}$$

Question 2

In Fig.1, a rod PR is constrained by guides to move horizontally and is driven by a crank OA and a sliding block at P. For the given configuration determine graphically or otherwise the acceleration of PR when OA has an angular velocity of 5 rad/s in a counter-clockwise direction and an angular acceleration of -35 rad/s^2 , i.e. Clockwise. Also, find the torque required at the driving shaft through O to overcome a horizontal force of 50 N along PR.



* Given:

$$\omega_{OA} = 5 \text{ rad/s } \curvearrowright$$

$$\alpha_{OA} = 35 \text{ rad/s}^2 \curvearrowleft$$

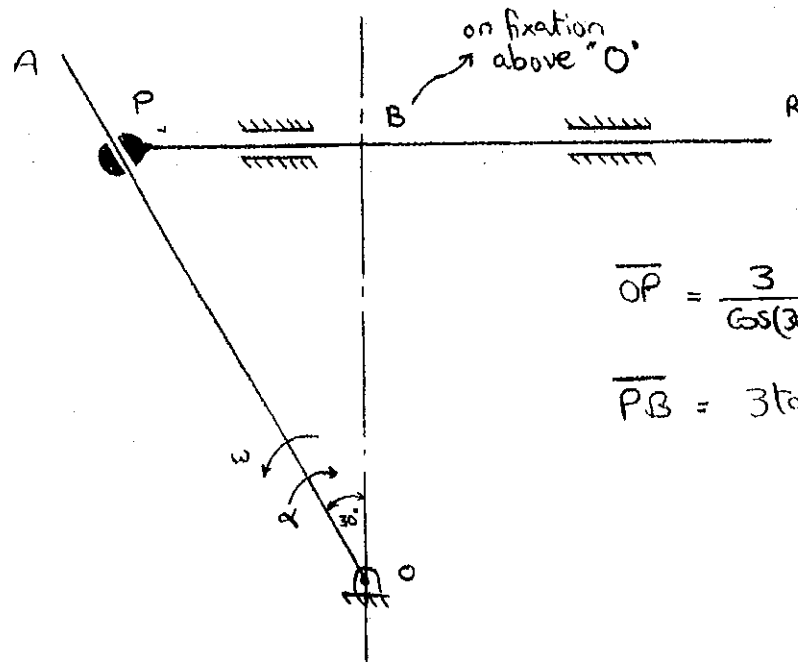
$$F = 50 \text{ N } \rightarrow$$

* Required:

$$- a_P = ?$$

$$- T_{OA} = ?$$

Cont. Q 2:



$$\overline{OP} = \frac{3}{\cos(30)} = 2\sqrt{3} \text{ m}$$

$$\overline{PB} = 3 \tan(30) = \sqrt{3} \text{ m}$$

Velocity Analysis:

* Loop OPBO;

$$\vec{V}_{PO} + \vec{V}_{BP} + \vec{V}_{OB} = 0.$$

$$\overline{OP} \omega_{OA} \angle 120+90 + V_{PO}^s \angle 120 + V_{BP}^s \angle 0 = 0$$

$$2\sqrt{3}(5) \angle 210 + V_{PO}^s \angle 120 + V_{BP}^s \angle 0 = 0$$

Im parts:

$$2\sqrt{3}(5) \sin(210) + V_{PO}^s \sin(120) + 0 = 0.$$

$$\Rightarrow V_{PO}^s = 10 \text{ m/s}$$

Re parts:

$$2\sqrt{3}(5) \cos(210) + 10 \cos(120) + V_{BP}^s \cos(0) = 0$$

$$\Rightarrow V_{BP}^s = 20 \text{ m/s}$$

